



ISP ENVIRONMENTAL SOLUTIONS PVT. LTD

RVM MACHINE USER MANUAL & SOP

Comprehensive Reverse Vending Machine Kiosk Manual, Screen-by-Screen Citizen Walkthrough with Real-Time Snapshots, and Technical Standard Operating Procedures (SOP)

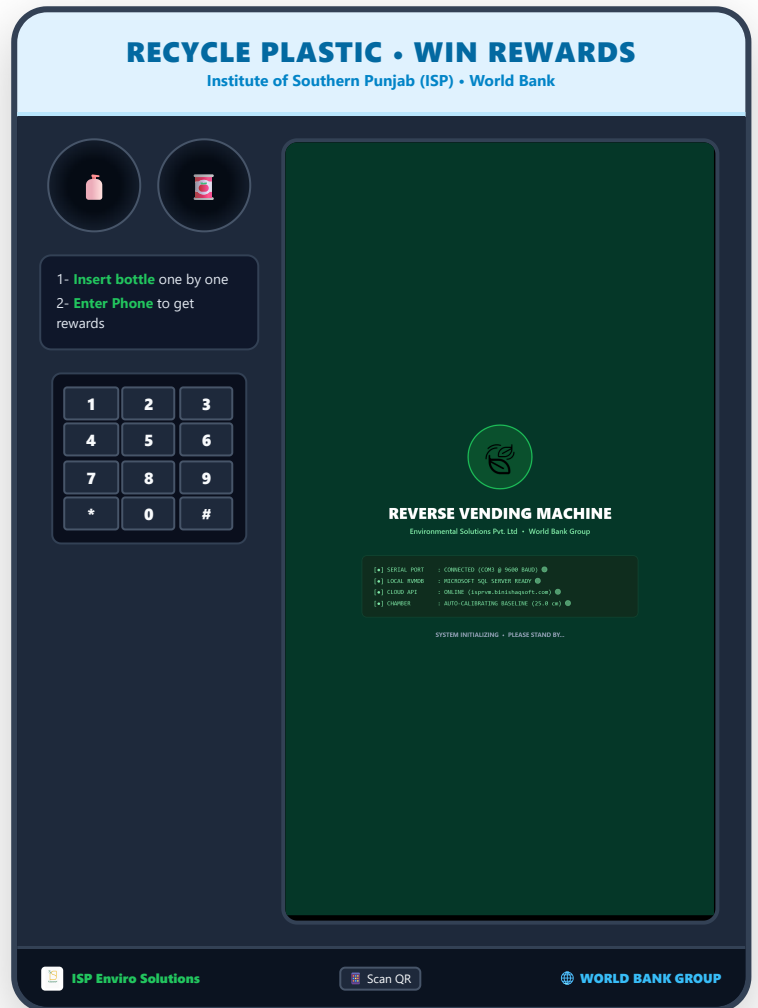
Operating Organization:	ISP Environmental Solutions Pvt. Ltd
Project Initiative:	EcoDrop Reverse Vending Kiosk Network
Partner Organization:	World Bank Group — Sustainable Circular Economy
Software System:	RVMDesktopApp (.NET 8.0 WPF / Arduino Uno)
Screen Captures:	Actual High-Resolution WPF Application Snapshots
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Splash Screen

شروعاتی اسکرین اور بارڈویئر ٹیسٹ

This screen indicates that the Reverse Vending Machine (RVM) is starting up and initializing subsystems. The system automatically tests the Arduino serial COM port, verifies local SQL database connectivity, connects to the Central Master Cloud, and calibrates the ultrasonic chamber distance.



ACTION REQUIRED:

Citizen Action: No action required. Stand by while the machine auto-calibrates (3–5 seconds). The Home Screen will appear automatically.

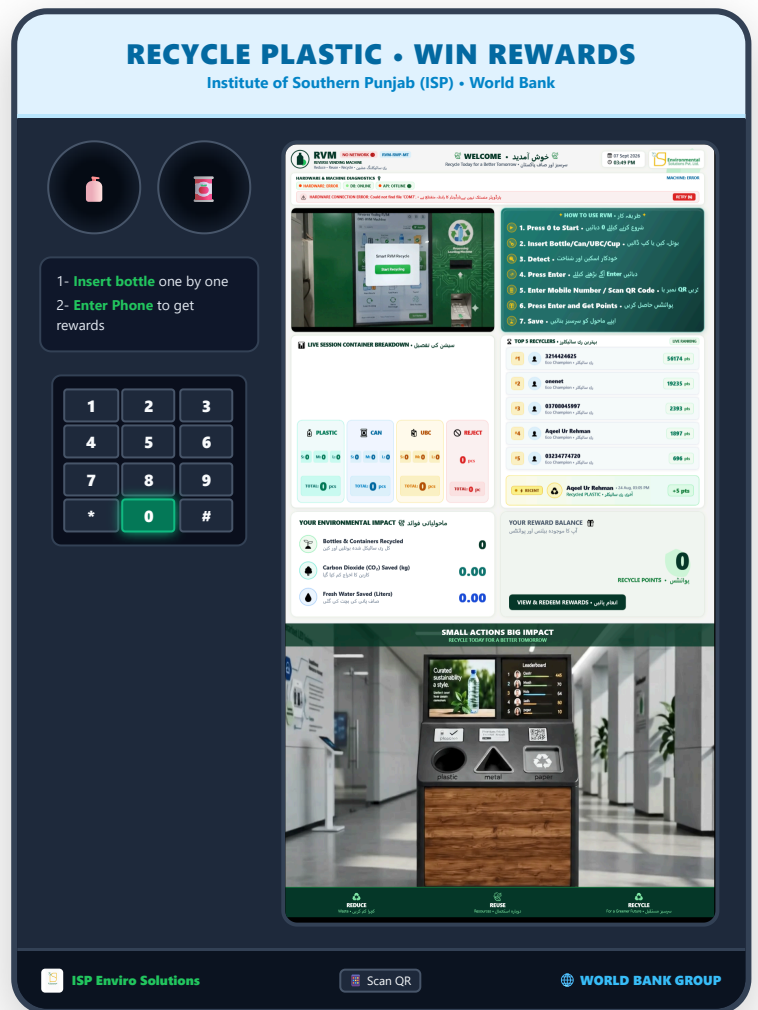
Serial Port:	COM3 @ 9600 Baud (Auto-Detect)
Local Storage:	Microsoft SQL Server RVMDb
Cloud Telemetry:	HTTPS isprvm.binishaqsoft.com
Chamber Baseline:	25.0 cm ± 2.0 cm (Ultrasonic)



Home Page

مرکزی اسکرین اور طریقہ کار

This is the primary standby welcome screen. Citizens are greeted with bilingual Urdu/English environmental messaging, live container intake counts, today's top community eco-champions, and cumulative environmental impact meters (CO2 & Water saved).



ACTION REQUIRED:

Citizen Action: Press the '*' or '0' button on the physical keypad, OR touch the glowing green circle on screen to start recycling.

Keypad Trigger: **Press '0' or '*' Key to Start**

Touchscreen: **Tap Central 310px Circular Button**

Bilingual Display: **English + Urdu Nastaliq Type**

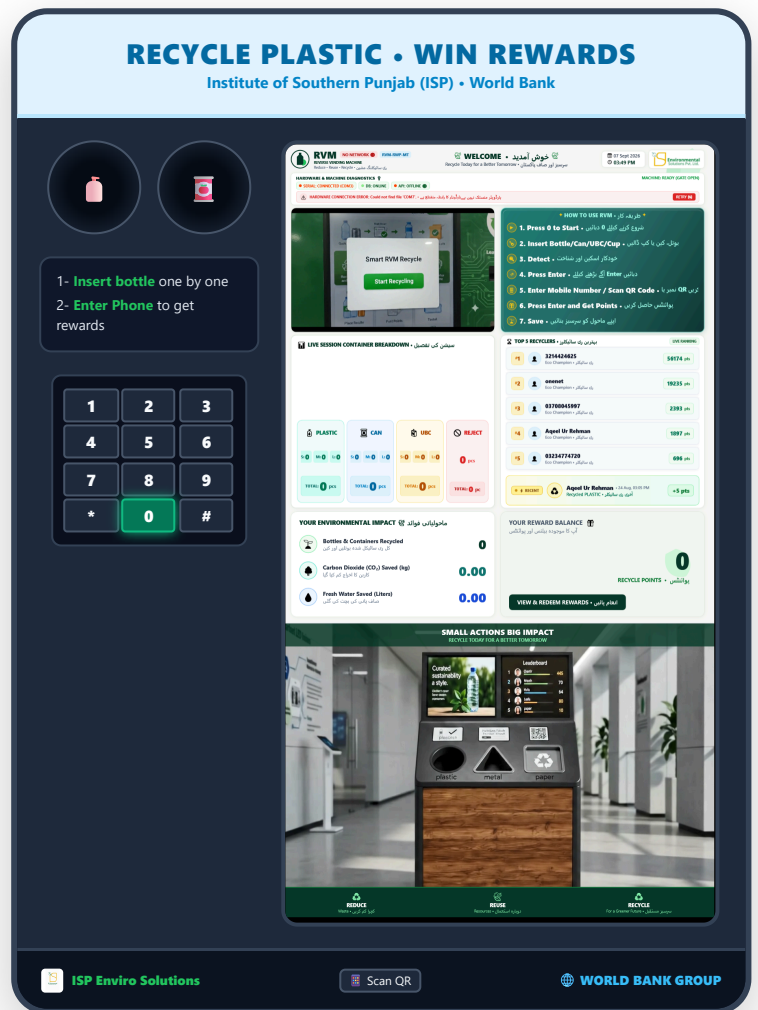
Community Board: **Top 5 Recyclers + Live Pulse**



Step: 01

پہلا مرحلہ: بوتل ڈالنے کی تیاری

This is the first active step of the recycling process. By pressing the '0' button or touching the screen, the citizen indicates readiness. The motorized security gate opens, internal chamber lights turn ON, and optical sensors prepare for container entry.



ACTION REQUIRED:

Citizen Action: Ensure container is empty of liquid. Insert bottle or can bottom-first into the open circular aperture.

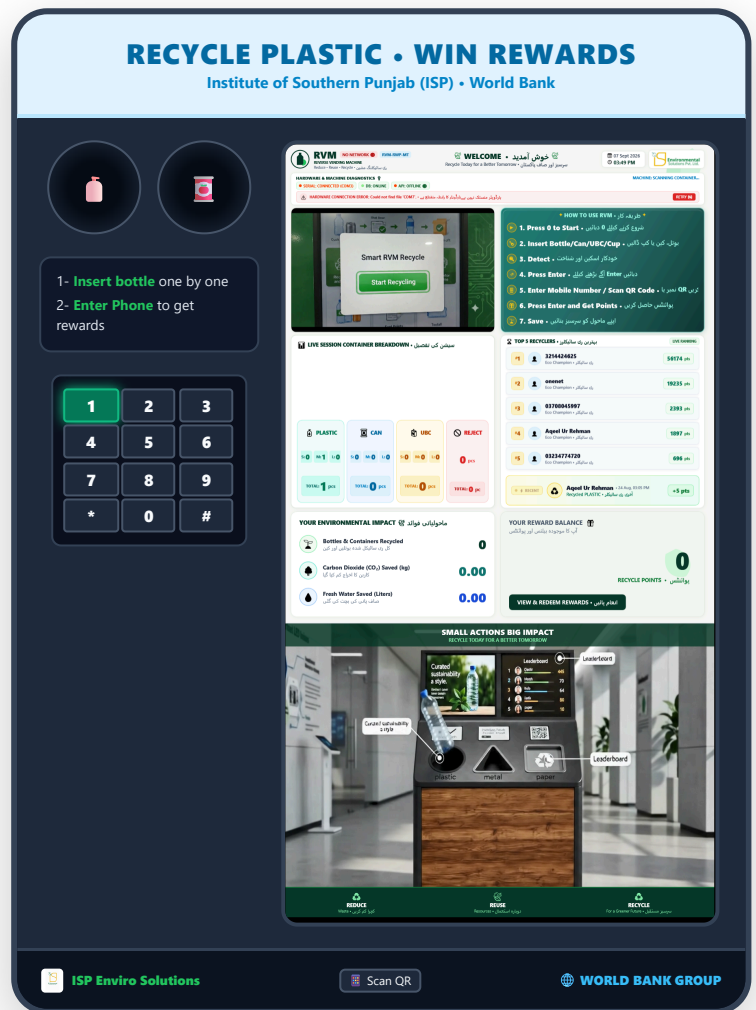
Intake Gate Servo:	Rotates to 90° (Aperture Open)
Chamber LED Ring:	Pin D13 HIGH (Bright White)
Inactivity Timer:	30-Second Citizen Timeout
Optical Baseline:	Ultrasonic Distance Armed



Step: 02

دوسرا مرحلہ: اسکیپنگ اور پیمائش

The RVM is actively detecting and classifying the inserted item. The multi-sensor array reads container height via 3-tier IR beams, tests for metal conductivity via the inductive sensor, and measures exact drop length via ultrasonic acoustic bounce.



ACTION REQUIRED:

Citizen Action: Allow container to settle inside chamber.
Press '1' on keypad or wait for automatic confirmation.

Height Sizing (IR): **Bottom (D2), Mid (D7), Top (D8)**

Inductive Metal (D5): **Metal Can (HIGH) / Plastic (LOW)**

Ultrasonic Cones: **Pins D3/D4 (Length in cm)**

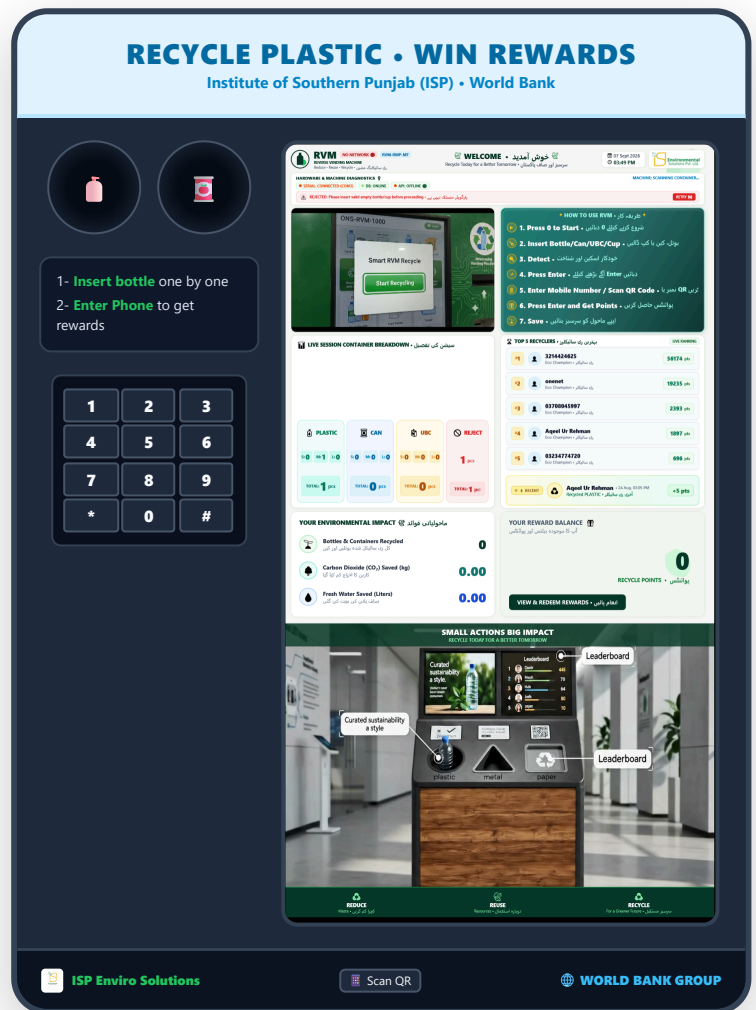
Classification Time: **< 850 ms Real-Time Response**



Step: 02 (Continue...)

نامنظور یا بوتل پھنس جانے کا انتباہ

If no bottle or cup is inserted, if non-recyclable garbage is detected, or if a bottle containing liquid is placed inside, the machine protects itself and displays a clear advisory alert, safely opening the gate for retrieval.



ACTION REQUIRED:

Citizen Action: Remove the rejected item from the aperture.
Insert an empty and acceptable plastic bottle or can.

Liquid Rejection: **Optical refraction detects residual liquid**

Anti-Pinch Safety: **Servo gate reverses upon resistance**

Jam Timeout: **Triggers after 4.0s blocked beam**

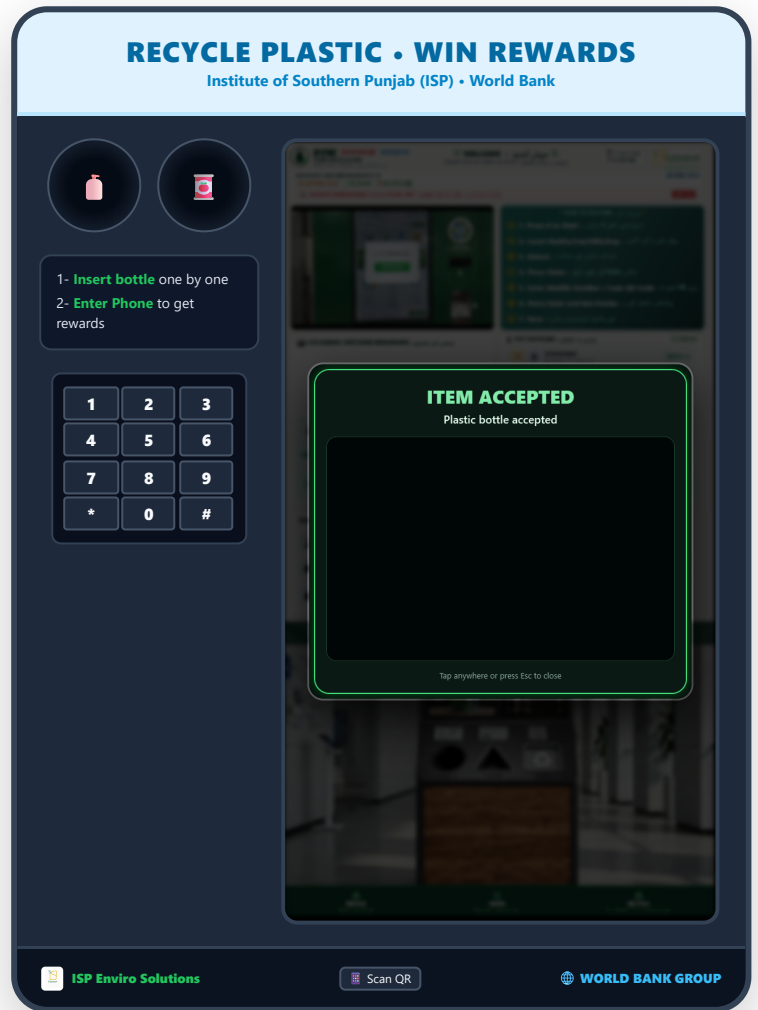
Audit Counter: **Increments RejectedTotalCount**



Step: 03

تیسرا مرحلہ: بوتل قبول اور پوائنٹس کا اندراج

Once the insertion is complete and verified, the servo drop gate drops the item into the collection bin. The celebratory modal triggers, reward points are allocated, and live environmental metrics increment immediately.



ACTION REQUIRED:

Citizen Action: Insert another container for more points, OR press 'Enter' on keypad to claim your rewards.

Drop Confirmation: **BOTTLE:CLEARED Ping Received**

Point Rules: **+5 S, +10 M, +15 L Points**

CO2 Ratio: **+0.15 kg CO2 Saved per container**

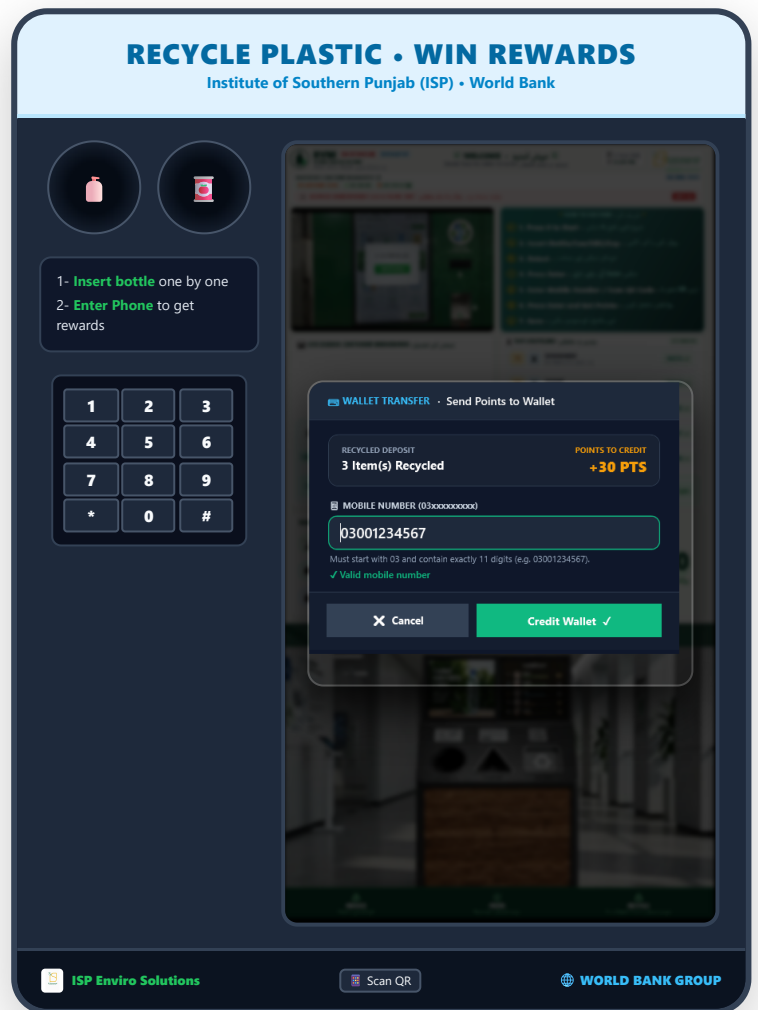
Water Ratio: **+0.75 L Fresh Water Saved**



Step: 04

چوتھا مرحلہ: انعام کیلئے موبائل نمبر کا اندراج

This screen displays the total earned points. The citizen enters their 11-digit Pakistani mobile phone number (03xxxxxxx) or scans their personal EcoDrop mobile app QR code to link points to their digital wallet.



ACTION REQUIRED:

Citizen Action: Enter your 11-digit phone number using the physical keypad or screen, then press 'Enter' or tap 'Credit Wallet'.

Mobile Format: **03xxxxxxx (Standard 11 Digits)**

Validation Regex: **^03[0-9]{9}\$ (Strict Digit-Only)**

Local Ledger: **dbo.WalletAccounts Account Commit**

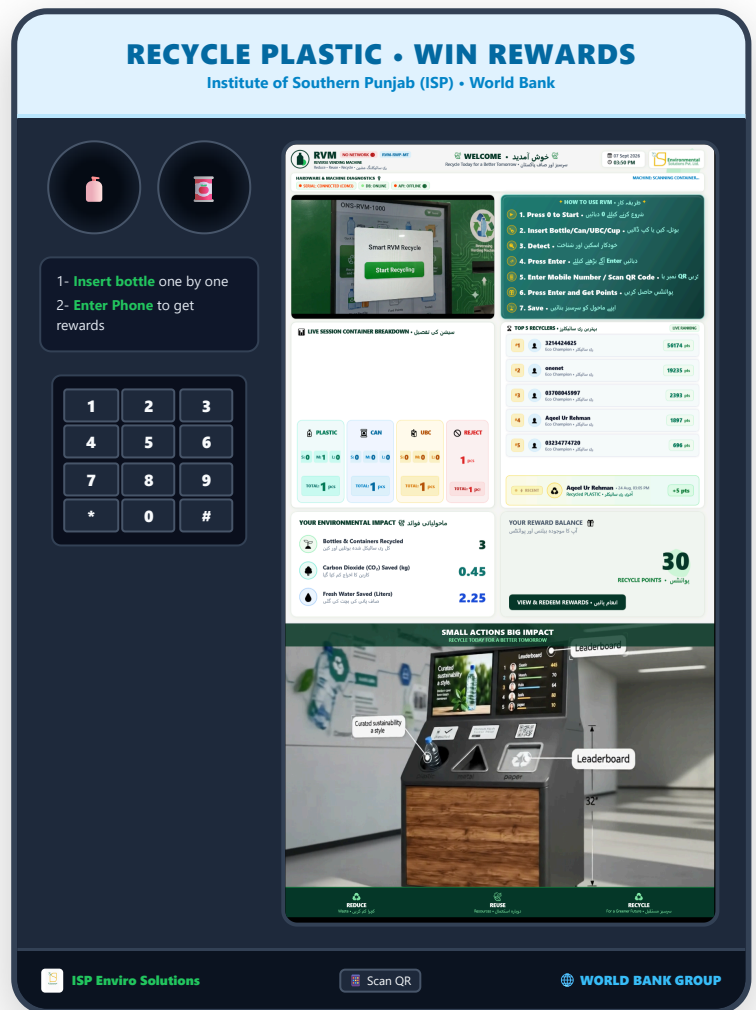
Alternative: **Personal QR Code Scan via Camera**



Step: 04 (Continue...)

کامیابی: پوائنٹس اکاؤنٹ میں منتقل ہو گئے

This screen indicates that the session has been successfully completed. Points are written to the local database, synced via secure API to the Central Cloud, and a confirmation receipt is presented.



ACTION REQUIRED:

Citizen Action: Review your credit confirmation and updated points balance. Proceed to customer feedback.

Cloud Endpoint: **POST /api/machine/sync-session**

Local Record: **dbo.Transactions Inserted**

Sync Flag: **IsSynced = 1 (Real-Time Cloud)**

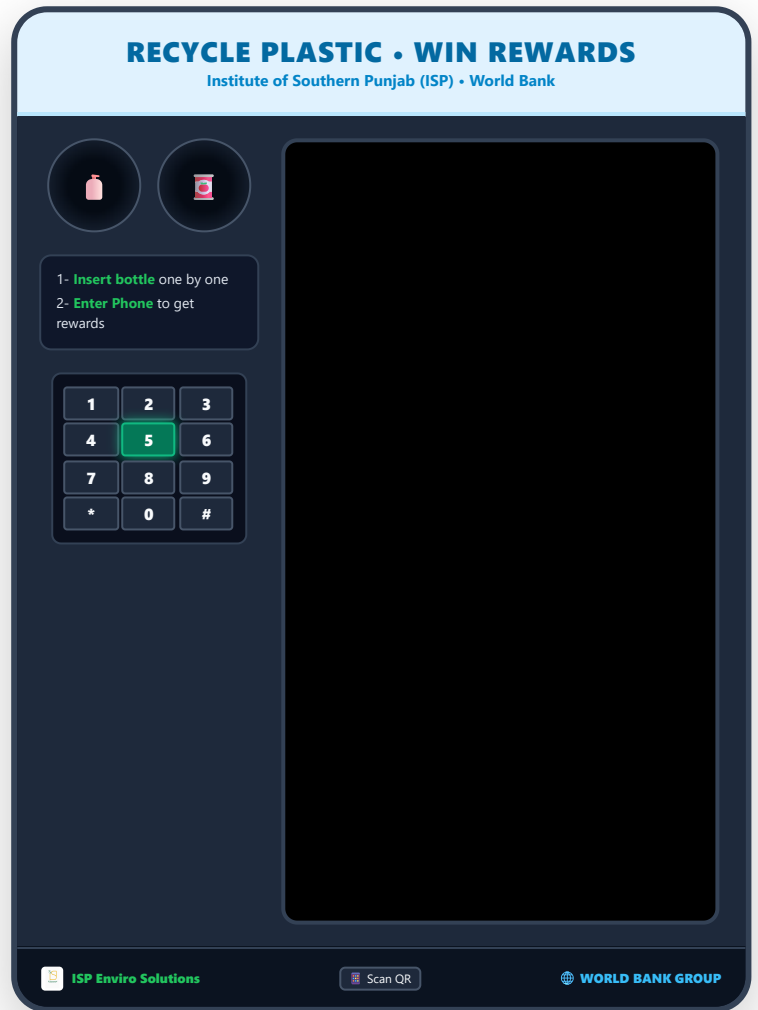
SMS Dispatch: **Dispatched to citizen phone**



Feedback

تجربے کی درجہ بندی اور فیڈبیک

On this screen, the citizen is prompted to provide their feedback regarding machine cleanliness, intake speed, and satisfaction using an interactive 5-star rating system.



ACTION REQUIRED:

Citizen Action: Press a keypad number from 1 to 5 (1 = Poor, 5 = Excellent) or tap the stars on screen.

Input Options: **Keypad Keys '1' to '5'**

Auto-Skip: **10-Second Inactivity Timeout**

Data Sync: **Uploaded to QA Central Dashboard**

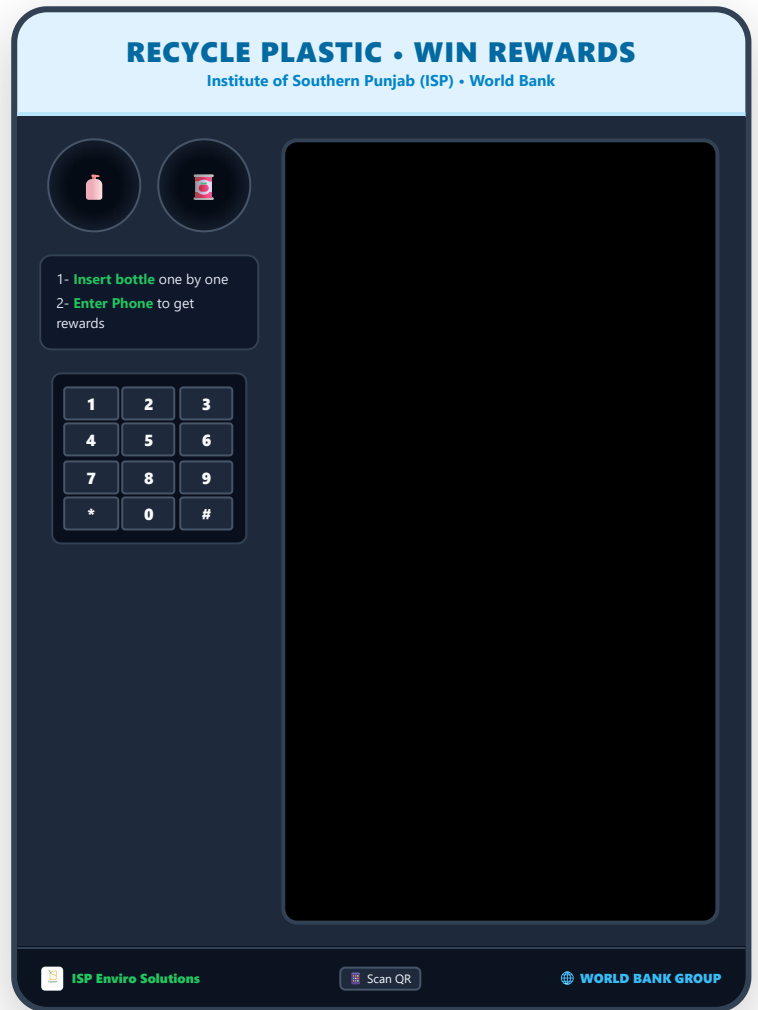
Metric: **Citizen CSAT Satisfaction Rating**



Process Completed

عمل مکمل اور مشین کی بحالی

The final acknowledgement screen thanking the citizen for their contribution to a clean and green environment. An automated 5-second countdown timer safely resets the machine for the next user.



ACTION REQUIRED:

Citizen Action: Process is complete! Enjoy your rewards and continue recycling for a cleaner Pakistan.

Countdown Timer: **5 Seconds Auto-Reset**

Gate Security: **Aperture locked in closed state**

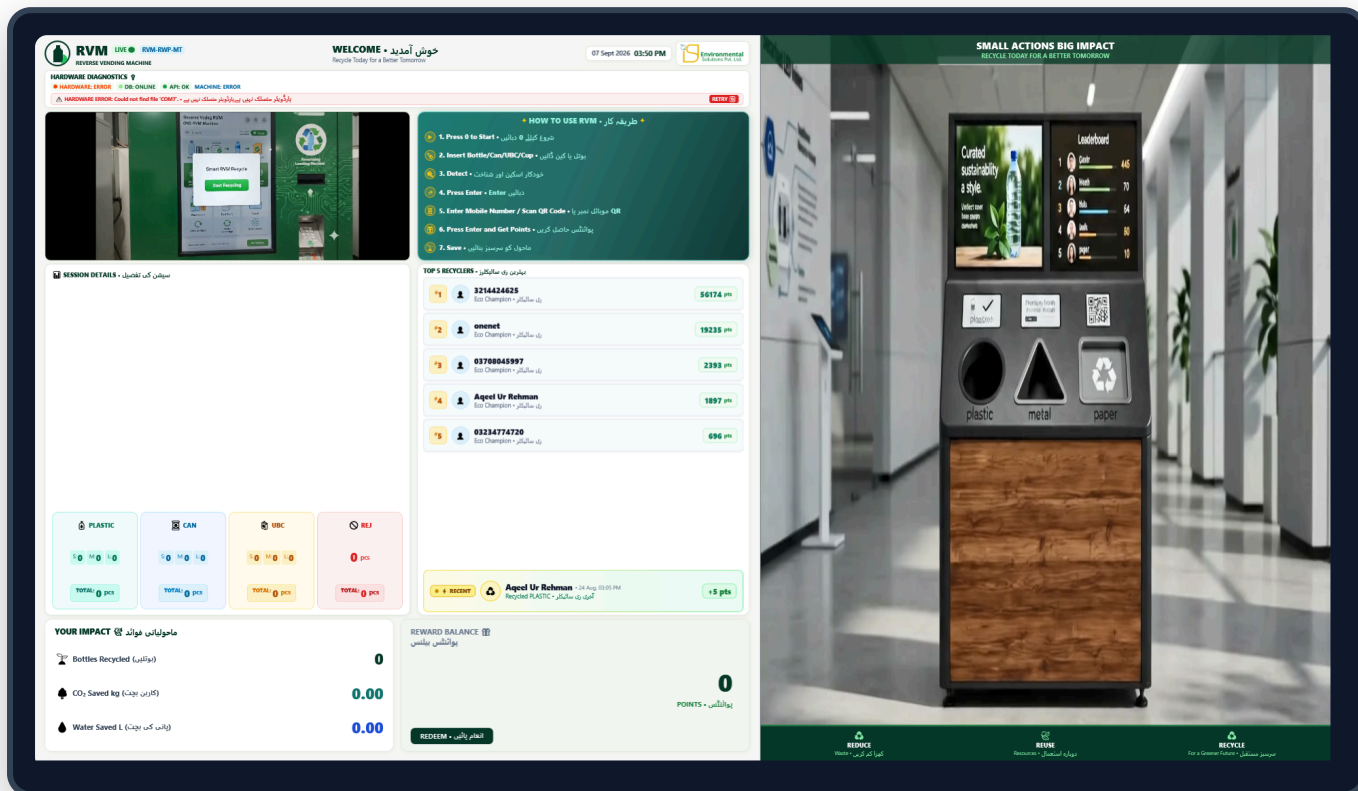
Session Buffer: **Flushed and ready for next user**

Signage Player: **Resumes commercial Ads loop**



LANDSCAPE KIOSK LAYOUT — RVMDesktopApp (1920 × 1080 Full HD)

Optimized Horizontal Display: Interactive Kiosk Dashboard (Left 56%) + Digital Video Signage (Right 44%)



LEFT HALF (56%): INTERACTIVE TOUCH KIOSK

- Full bilingual English + Urdu Nastaliq Header with live machine diagnostics.
- Circular "Insert Bottle/Can/UBC" start action button with live counter updates.
- Live Session Breakdown: Plastic, Aluminium Can, UBC Cartons, and Rejected items.
- Teal Glassmorphic 7-Step guide and Top 5 Recyclers community leaderboard.

RIGHT HALF (44%): DIGITAL SIGNAGE & ADS

- Continuous commercial video advertising loop in high-definition 1080p.
- Top hero signage banner: "Small Actions Big Impact — Recycle Today".
- Bottom eco sustainability footer: Reduce, Reuse, Recycle awareness slogans.
- Hidden slide-out real-time serial telemetry console accessible via hotkey '8'.



STANDARD OPERATING PROCEDURES — PART 1 (OPERATIONS & MAINTENANCE)

Official Field Guidelines for Caretakers, Municipal Handlers & Technicians

SOP-RVM-OPS-01

Daily 07:00 AM

Daily Morning Startup & Diagnostics

- ✓ Inspect AC 220V UPS connection and power on master rocker switch.
- ✓ Confirm Windows IoT executes `launch-kiosk.ps1` in portrait mode on Screen 1.
- ✓ Verify Diagnostics: `SERIAL: CONNECTED` ●, `DB: OK` ●, `API: ONLINE` ●.
- ✓ Execute test cycle: Insert test container and verify +10 point allocation.
- ✓ Log pre-flight sign-off in physical door register.

SOP-RVM-OPS-02

Continuous Public

Citizen Operation & Material Rules

- ✓ Accepted: Clear & colored PET bottles, Aluminium UBC Cans, Paper cups.
- ✓ Prohibited: Glass, chemical/oil containers, filled liquids, crushed cans.
- ✓ Enforce single container feed: Insert one-by-one, base-first.
- ✓ Validate Pakistani mobile phone numbers (`03xxxxxxxx` - 11 digits).
- ✓ Assist citizens with QR code mobile app onboarding and redemption.

SOP-RVM-MAINT-03

At 85% Capacity

Storage Bin Emptying & Sensor Reset

- ✓ Type `888` + Admin PIN on keypad to pause active sessions.
- ✓ Unlock bottom cabinet with master key; extract full 120L trolley liner.
- ✓ Wipe dust from Pin D10 (Plastic IR) and D11 (Can IR) level sensor lenses.
- ✓ Insert fresh heavy-duty transparent collection liner; lock cabinet door.
- ✓ In Admin Panel Tab 4, click "Reset Bin Full Counter".

SOP-RVM-MAINT-04

Weekly (Friday)

Chamber Cleaning & Calibration

- ✓ Wipe Bottom (D2), Mid (D7), Top (D8) IR lenses using 90%+ IPA wipes.
- ✓ Wipe inductive metal faceplate (D5); clean dust from ultrasonic cones.
- ✓ Lubricate servo drop flap pivot hinge with dry silicone spray.
- ✓ Press `33` on physical keypad to execute 20-ping baseline calibration.
- ✓ Verify standard 500ml PET bottle registers correctly.



STANDARD OPERATING PROCEDURES — PART 2 (SYSTEM & SAFETY)

Network Resiliency, Emergency Protocols, Signage & Security

SOP-RVM-NET-05

Network Failover

Offline Resiliency & Cloud Sync

- ✓ When 4G drops, system stays 100% operational in offline caching mode.
- ✓ Transactions commit locally to SQL Server with flag 'IsSynced = 0'.
- ✓ Upon network recovery, background service auto-pushes FIFO batches.
- ✓ Technician force override: Admin Window Tab 3 -> "Sync Pending".
- ✓ Confirm pending records drop to 0 after sync completes.

SOP-RVM-SAFE-06

Emergency Jam

Emergency Jam Clearance & Safety

- ✓ PINCH HAZARD: Immediately press Red Mushroom Emergency Stop button.
- ✓ Unlock upper chamber service hatch using physical maintenance key.
- ✓ Using rubberized tongs or gloves, extract stuck container or debris.
- ✓ Inspect sensor optical pathways for scratches or physical damage.
- ✓ Twist and release Emergency Stop button clockwise to reboot system.

SOP-RVM-MKT-07

Weekly Media

Digital Signage Video Ads Management

- ✓ Standard Format: MP4 (H.264 / AAC), 1080x720 or 1080x1920, 25/30 fps.
- ✓ Copy approved promotional media files into 'C:\RVM\Ads\' directory.
- ✓ Open Admin Window ('888') -> Tab 6 (Advertisement & Video Signage).
- ✓ Select active videos, verify preview playback, and save playlist.
- ✓ Confirm seamless loop transitions on secondary signage display.

SOP-RVM-ADMIN-08

As Needed

Technician Hotkeys & Threshold Tuning

- ✓ '888': Open Admin Security PIN login prompt.
- ✓ '111': Direct service mode bypass for field debugging.
- ✓ '33': Trigger instant 20-sample chamber acoustic recalibration.
- ✓ '8': Toggle hidden real-time serial telemetry console terminal.
- ✓ Adjust container sizing thresholds inside root 'config.txt' file.



ARDUINO UNO PINOUT & RAPID TROUBLESHOOTING GUIDE

Industrial Pin Assignments & Field Corrective Action Matrix

Pin	Direction	Hardware Subsystem	Operational Logic & Pin Function
D2	INPUT	Bottom IR Photoelectric Sensor	Detects container entrance; triggers classification timer.
D3	OUTPUT	Ultrasonic Distance Sensor (Trig)	Emits 40 kHz acoustic burst to gauge container drop speed & height.
D4	INPUT	Ultrasonic Distance Sensor (Echo)	Measures reflected acoustic return pulse in microseconds.
D5	INPUT	Inductive Proximity Sensor	HIGH = Aluminium Can / Metal. LOW = PET Plastic / Paper.
D7	INPUT	Middle IR Photoelectric Sensor	Height trigger for Medium containers (> 18 cm threshold).
D8	INPUT	Top IR Photoelectric Sensor	Height trigger for Large bottles (> 26 cm threshold).
D9	OUTPUT	Drop Gate Servo Motor (PWM)	0° = Closed Security Gate, 90° = Drop Open into collection bin.
D10	INPUT	Plastic Bin Level Optical Sensor	Infrared reflective sensor at rim. LOW = Storage Bin Full Alert.
D11	INPUT	Can/Metal Bin Level Sensor	Infrared beam at metal bin rim. LOW = Storage Bin Full Alert.
D13	OUTPUT	Internal Chamber LED Ring	Illuminates intake scanning chamber during active deposits.

RAPID FAULT RESOLUTION MATRIX

Symptom / Fault Code	Probable Root Cause	Immediate Corrective Action
SERIAL: DISCONNECTED  "HARDWARE CONNECTION ERROR"	USB cable dislodged; Arduino unpowered; incorrect COM port in `config.txt`.	1. Re-seat USB cable. 2. Verify COM port in Device Manager. 3. Click RETRY  .
API: OFFLINE  Yellow telemetry indicator	4G router outage; SIM data expired; DNS resolution timeout.	1. System auto-switches to offline mode; recycling continues. 2. Auto-syncs upon reconnection.
BIN FULL ALERT  "Storage Bin Full - Call Operator"	120L storage trolley at capacity; dust covering D10/D11 optical lenses.	1. Follow SOP-03. 2. Replace collection liner. 3. Wipe lenses with IPA. 4. Reset counter in Tab 4.
BOTTLE STUCK / JAM  "Bottle stuck - Remove container"	Oversized or bent container; servo gate obstruction; beam blocked > 4s.	1. Follow SOP-06. 2. Press Emergency Stop. 3. Clear chamber with safety tongs.
INVALID PHONE NUMBER  "Must be 11 digits (03...)"	User entered landline, incomplete digits, or non-numeric characters.	1. Ensure number begins with 03. 2. Exactly 11 digits required (e.g. 03001234567).